

Ghazal Lahooti

Urban Designer & Researcher

Data-driven urban design · Early-stage climate & comfort analysis · Custom tool development

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Portfolio · *Click or Scan*

Profile

Urban designer and researcher working on data-driven design for the earliest phases of a project. I specialise in urban heat and nature-based solutions (NBS): which streets to cool first, based on where heat meets the places people walk every day, and which NBS each street can hold. I work with satellite data, GIS, sun and wind simulation and parametric tools, and build my own tools when existing ones fall short, most recently UrbanSim, a browser-based heat-screening tool for planners.

Core tools: Ladybug · Eddy3D & SimScale · Grasshopper & Galapagos · QGIS · Google Earth Engine · Python
Methods: remote sensing, sun & wind simulation, parametric optimisation, systematic & umbrella reviews

Education

- **M.Sc. in Sustainable Urban Design**, Lund University (LTH), Sweden Sep. 2024 – Oct. 2026
  Thesis: *A Context-Specific Nature-Based Framework for Heat Mitigation: Supporting the Everyday Mobility of Older Adults in Kadıköy, Istanbul*
  Supervisor: Teresa Arana Aristi · Examiner: Andrew Karvonen
- **B.Sc. in Architectural Engineering**, Iran University of Science and Technology, Iran Sep. 2018 – Jul. 2023
  GPA: 16.95/20 · Thesis: *Sustainable Mountain Refuge at the Third Campsite of Mount Damavand* (passive climate strategies for an extreme environment)

Research & Development Projects

- **Heat Mitigation Framework for Older Adults' Everyday Mobility** (M.Sc. Thesis), Istanbul Jan. – Oct. 2026
  Lund University · Fieldwork funded by the Crafoord Foundation (SEK 30,000)
  How can NBS cool older adults' everyday walks in Kadıköy, Istanbul's district with the most residents 65+?
  - Mapped ten-summer surface temperature (Landsat, Earth Engine) against residents aged 65+ in QGIS.
  - Linked 124 destinations into the *Green Loop*, a cooling network within 100 m of two-thirds of them.
  - Grouped 10 streets into 4 typologies using Ladybug sun simulation; designed with a 7-tool NBS toolbox.
  - **Outcome:** a transferable framework: social context decides *where* to cool, physical context decides *how*.
- **UrbanSim: Heat-Screening Tool for NBS** (Founder & sole developer) Summer 2026 – present
  Sustainability Leapfrogs, LU Innovation, Lund University · SEK 40,000 scholarship
  A browser-based tool showing planners where heat meets everyday life and which NBS fit each street.
  - Ranks street segments by pedestrian sun exposure, nearby destinations and surface heat, with live weights.
  - Matches NBS through traceable, source-cited rules; JavaScript, OpenStreetMap, Landsat; GIS export.
  - **Status:** conceptual MVP. Next phase: build an evidence-graded NBS library with academic partners and train a model on it to recommend suitable NBS for each street.

Research Experience

- **Research Intern**, Institute for Design Strategies, TH OWL, Detmold, Germany Oct. 2026 – Dec. 2026
  AIM-NBS: Advancing Impact Measurement of Nature-Based Solutions (second phase)
  - Recalibrating the umbrella review findings for the project's final synthesis.
  - Writing the project report, in which I will be credited.
- **Project Assistant**, Lund University & City of Malmö Oct. 2025 – Nov. 2025
  Microclimate analysis of Hyllie, Malmö, Sweden
  - Rebuilt and optimised the 3D site model for CFD wind simulation (run by Erik Johansson, Lund University).
  - Mapped direct sun hours per grid cell to identify the least sun-exposed areas.
- **Research Intern**, Institute for Design Strategies, TH OWL, Detmold, Germany Jun. 2025 – Aug. 2025
  AIM-NBS: Advancing Impact Measurement of Nature-Based Solutions (first phase)
  - Covered 6 of 12 societal challenges in a PRISMA umbrella review of NBS impacts; analysed 410 papers.
  - Extracted and synthesised qualitative and quantitative impact indicators for those six challenges.
  - Visualised the results in Python (Pandas, Matplotlib).

Professional Experience

- **Junior Architect**, Cedrus Architecture Studio, Tehran Feb. 2023 – Aug. 2024
  Preceded by internship, Jan. – Feb. 2023
  - Site and environmental analyses and precedent research for early-stage housing and building design.
  - Communicated design proposals through presentation documents, visualisations and animations.
  - Designed residential interiors and built physical models.

Research Funding & Awards

- **Sustainability Leapfrogs Scholarship**, LU Innovation (SEK 40,000), 2026: development of UrbanSim
- **Crafoord Foundation Travel Grant** (SEK 30,000): thesis fieldwork in Istanbul
- **Lund University Global Scholarship** (75% tuition waiver): awarded for academic merit

Skills & Methods

- **Research methods:** PRISMA systematic & umbrella reviews, interviews, street surveys, demographic analysis
- **Spatial data:** QGIS, Google Earth Engine (Landsat), Python (GeoPandas, Pandas, NumPy, Matplotlib)
- **Simulation & optimisation:** Ladybug, Eddy3D, SimScale, Galapagos (evolutionary optimisation)
- **Tools & automation:** Python, OpenStreetMap, GeoJSON, Grasshopper components
- **Design & modelling:** Rhino, Grasshopper, Revit, AutoCAD, SketchUp, Blender
- **Visual communication:** InDesign, Illustrator, Photoshop, D5 Render, Enscape, After Effects

Languages

- **Persian**: Native
- **English**: C1 (IELTS Academic 7.5)
- **German**: B1 (Goethe-Zertifikat)
- **Swedish**: A1, currently learning